

G-space and equiv homotopy groups

background for the Hill-Hopkins-Ravenel Kervaire invariant One Thm.

Def (homotopical category)

A homotopical category is a pair $(\mathcal{C}, \mathcal{W})$, where $\mathcal{C}$ is a category, and $\mathcal{W}$ is a collection of morphisms in $\mathcal{C}$ called weak equivalences. $\mathcal{W}$ satisfies the following

- contains all identity maps
- satisfies "2 out of 6 property" that if vu and wv are in $\mathcal{W}$, then u, v, w, vwu are also in $\mathcal{W}$.

$$\bullet \xrightarrow{u} \bullet \xrightarrow{v} \bullet \xrightarrow{w} \bullet$$

Ex

$(\text{Top}, \mathcal{W})$

• Top: the category consisting with top spaces and conti maps

• $\mathcal{W}$: all weak homotopy equivalences

$\hookrightarrow f: X \rightarrow Y, f$ induces

$$\pi_0(X) \xrightarrow{\cong} \pi_0(Y), \pi_n(X, x) \xrightarrow{\cong} \pi_n(Y, y), \forall n \geq 1.$$

For $X \xrightarrow{u} Y \xrightarrow{v} Z \xrightarrow{w} T, vu, wv \in \mathcal{W}$. So in

$$\pi_n(X) \xrightarrow{u_*} \pi_n(Y) \xrightarrow{v_*} \pi_n(Z) \xrightarrow{w_*} \pi_n(T),$$

(1)

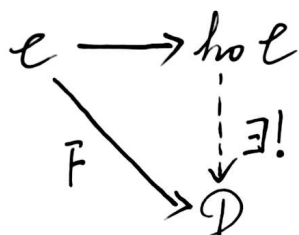
v_*u_* and w_*v_* are isos. So u_* and v_* are injections, and v_* and w_*

are surjections. So u_* is an iso. Then, u_* , w_* and $w_* u_* u_*$ are also isos. So $(\text{Top}, \mathcal{W})$ is a homotopical category.

Def (homotopy category)

For a homotopical category $(\mathcal{C}, \mathcal{W})$, the corresponding homotopy category $ho\mathcal{C}$ with the functor $\mathcal{C} \rightarrow ho\mathcal{C}$ satisfy that

for every category $\mathcal{D}$ and every functor $F: \mathcal{C} \rightarrow \mathcal{D}$ taking weak equivalences as isos, $\exists!$ functor $ho\mathcal{C} \rightarrow \mathcal{D}$, s.t. the diagram commute.



of G :

(2)

G is a finite group. $\mathcal{T}^G$ is a category consisting of

- objects: pointed left G -space (X, x_0) , where X is compactly generated, weak Hausdorff, with conti left action $G \times X \rightarrow X, (g, x) \mapsto gx$. $gx_0 = x_0$ for $\forall g \in G$.

- morphisms: $f: X \rightarrow Y$ is G -equiv, where $f(x_0) = y_0$ and $f(gx) = gf(x)$, for $g \in G, x \in X$.

$\mathcal{T}^G(X, Y)$ is a top space consisting of equiv map from X to Y .

So $\mathcal{T}^G$ is a top category. G acts on $X \wedge Y$, smash product, as $g \cdot (x \wedge y) = (gx) \wedge (gy)$. $\mathcal{T}^G$ is a closed symmetric monoidal category under the smash product operation $\wedge$.

$$X \vee Y = (X \times \{y_0\}) \cup (\{x_0\} \times Y) \subset X \times Y$$

$$X \wedge Y := X \times Y / X \vee Y$$

And $S^0 = \{*, 1\}$ with trivial G -action is the tensor unit.

Def:

$(\mathcal{V}, \otimes, 1)$ is a monoidal category. A category $(\mathcal{C})$ enriched over $\mathcal{V}$ consist of

- objects: a collection of objects $X, Y, Z, \dots$
- morphisms: For $X, Y \in \text{Ob}(\mathcal{C})$, $\mathcal{C}(X, Y) \in \mathcal{V}$, called the hom-object from X to Y . That means that the morphisms from X to Y form a object in $\mathcal{V}$.

Let $\mathcal{I}_G$ be a enriched category over $\mathcal{T}^G$. $\mathcal{I}_G$ is the G -equiv top category of G -spaces and G -spaces of coord.

The G -action on the mapping space is

$$(g \cdot f)(x) = g \cdot f(g^{-1} \cdot x).$$

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And the composition is the G -equiv morphism

$$\mathcal{I}_G(Y, Z) \wedge \mathcal{I}_G(X, Y) \rightarrow \mathcal{I}_G(X, Z)$$

given by $(h, f) \mapsto h \circ f$. We can have

$$\mathcal{I}_G(X, Y) = \mathcal{I}^G(X, Y).$$

Def:

G is a finite group. $H \leq G$. X is a pointed G -space with basepoint x_0 . For $n \geq 0$, give S^n the trivial H -action. Define

$$\pi_n^H(X) := [S^n, X]_*^H$$

to be the set of pointed H -equiv maps $f: S^n \rightarrow X$ modulo pointed H -equiv homotopy.

This is the same as the ordinary homotopy group $\pi_n(X^H)$ of the space of H fixed-points in X , where $X^H = \{x \in X \mid hx = x \text{ for } \forall h \in H\}$.

When we define $\pi_n^H(X)$, H -action is trivial on S^n . If $f: S^n \rightarrow X$ is H -equiv, then $f(s) = f(h \cdot s) = h \cdot f(s)$ for $\forall h \in H$. So $f(s)$ must be fixed-points. So $f(s) \in X^H$. So every H -equiv is a composite

$$S^n \rightarrow X^H \rightarrow X.$$

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On the other side, any ordinary pointed map $f: S^n \rightarrow X^H$ is H -equiv, since every point in X^H is fixed under H . So we have

$$\text{Map}_*^H(S^n, X) \cong \text{Map}_*(S^n, X^H).$$

Then, considering homotopy classes, we get

$$[S^n, X]_*^H \cong [S^n, X^H]_*,$$

which means $\pi_n^H(X) \cong \pi_n(X^H)$.

$f: X \rightarrow Y$ is a weak equivalence in $\mathcal{T}^G$ if for all $H \in G$, and all choices of base points $x_0 \in X^H$, the induced map $\pi_n^H(X, x_0) \rightarrow \pi_n^H(Y, f(x_0))$ is an iso. With the weak equivalences, $\mathcal{T}^G$ is a homotopical category.

When X is cofibrant and Y is fibrant, we can calculate $[X, Y]^G$ as follows:

$$[X, Y]^G = \pi_0 \mathcal{T}^G(X, Y) = \pi_0 \underline{\mathcal{T}}^G(X, Y)$$

cofibrant:

X is cofibrant, if

$$* \rightarrow X$$

is cofibration.

fibrant:

Y is fibrant, if

$$Y \rightarrow *$$

is fibration.

For commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{a} & E \\ \downarrow i & \wr & \downarrow p \\ B & \xrightarrow{b} & Y \end{array}$$

if $\exists L: B \rightarrow E$, s.t. $L \circ i = a$, $p \circ L = b$, we say this block has a lift. If for any such kind of block, lift exists, we say i has left lifting property for p , and p has right lifting property for i , denoted $i \perp p$.

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W is weak equivalence. $i: A \rightarrow B$ is called a cofibration, if $i \perp (Fib \cap W)$, meaning has left lifting property for all maps that is fibration and weak equivalence at the same time. That is

$$Cof = {}^\perp(Fib \cap W)$$

$p: E \rightarrow Y$ is called a fibration, if $(Cof \cap W) \perp p$, meaning it has right lifting property for all maps that is cofibration and weak equivalence at the same time. That is

$$Fib = (Cof \cap W)^\perp$$

Exap
Top

$W = \{ \text{weak homotopy equivalence} \}$

$Fib = \{ \text{Serre fibrations} \}$

$Cof = \{ \text{maps that have left lifting property for all trivial Serre fibrations} \}$

• $W: f: X \rightarrow Y$ is weak equivalence, if it induces isos

$$f_*: \pi_n(X, x) \xrightarrow{\sim} \pi_n(Y, f(x))$$

• $Fib: p: E \rightarrow B$ is Serre fibration, if it has right lifting property for

all $D^n \times \{0\} \hookrightarrow D^n \times I$.

$$\begin{array}{ccc} D^n \times \{0\} & \xrightarrow{\quad} & E \\ \downarrow & \nearrow & \downarrow p \\ D^n \times I & \xrightarrow{\quad} & B \end{array}$$

(6)

$F \times B \rightarrow B$ and $IR \rightarrow S^1$ (covering) are fibration. All top spaces are

fibrant, since $X \rightarrow *$ is Serre fibration.

• Cof: $i: A \rightarrow X$ is cofibration if $i \perp (\text{Fib} \cap \mathcal{W})$.

If X is a CW-complex and A is its subcomplex, the inclusion $A \hookrightarrow X$ is a cofibration.

V is a rep of G . S^V is a one point compactification of V , with $g \cdot \infty = \infty$. Write $S^{V+n} = S^{V \oplus \mathbb{R}^n}$.

Define $\pi_V^G X = [S^V, X]^G$ to be the set of homotopy classes of G -equiv maps from S^V to X . $\pi_V^G X$ is a group if $\dim V > 0$ and an abelian group if $\dim V^G > 1$, where V^G is the space of G -invariant vectors in V .

We can also define the equiv suspension $\Sigma^V X = S^V \wedge X$, and the equiv loop space $\Omega^V X = \mathcal{I}_G(S^V, X)$.

Next we consider two orthogonal reps of G , V_1 and V_2 , s.t.

$\downarrow$
a rep V is orthogonal, if

$$\langle g \cdot v, g \cdot w \rangle = \langle v, w \rangle, \forall v, w \in V, \forall g \in G. \quad (7)$$

for each irreducible rep U of G in V_1 , it satisfies

$$\dim \underline{\text{hom}}^G(U, V_2) \geq \dim \text{hom}^G(U, V_1).$$

$$\hookrightarrow \text{hom}^G(U, V) = \{f: U \rightarrow V \mid f \text{ is } \mathbb{R}\text{-linear}, f(gu) = g f(u), \forall g \in G, u \in U\}$$

Then, choose an embedding $t: V_1 \hookrightarrow V_2$, s.t. $t(gv) = gt(v)$ and $\langle t(v), t(w) \rangle_{V_2} = \langle v, w \rangle_{V_1}$. Denote $V_2 - t(V_1) := t(V_1)^\perp$. Then, $V_2 - t(V_1)$ is an orthogonal G -rep. So we can consider

$$\pi_{V_2 - t(V_1)}^G(X).$$

$O(V_1, V_2)^G$ is a top space consisting of all G -equiv linear isometric embedding from V_1 to V_2 . For $V \in O(V_1, V_2)^G$, it corresponds to a group $\pi_{V_2 - t(V_1)}^G(X)$.

Def:

Write $V_1 < V_2$, if for every irreducible G -rep U ,

$$\dim \text{hom}^G(U, V_1) < \dim \text{hom}^G(U, V_2) - 1$$

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If the embedding t moves along a path $t_s: V_1 \hookrightarrow V_2$, $s \in [0, 1]$, then we can carry $\pi_{V_2 - t_0(V_1)}^G(X)$ to $\pi_{V_2 - t_1(V_1)}^G(X)$ along this path.

This forms a local system of groups. If $V_1 > V_2$, $O(V_1, V_2)^G$ is simply connected. So local system has no non-trivial monodromy.

Then, we can define

$$\pi_{V_2 - V_1}^G(X) := \Gamma(O(V_1, V_2)^G, \mathbb{Z}) \quad \text{this is the local system mentioned above}$$

If V_1 is a trivial rep of dim k , we write $\pi_{V_2 - k}^G(X)$.